

Proof Audit Report

Topology-switchable finite-horizon response operators

Independent Codex blind review

18 July 2026

Mathematical proof verdict

PASS. Proposition 1, Corollary 1, the unpenalized FWL/design-identification statement, the joint-ridge Schur complement and Proposition 2 have no open mathematical defects under their stated hypotheses.

End-to-end manuscript status

FAIL. This proof verdict does not authorize submission. Corrected RCEP and NYC scientific outputs have not been regenerated, and the production helper trust evidence is absent. Reason code: `corrected_implementation_not_regenerated`.

Review protocol

The reviewer received only the main theory source and Supplementary Notes 1 and 3. Prior audits, empirical outputs and R006e/R006f outcomes were excluded. The review checked definitions, quantifiers, hypothesis discharge, inequalities, interchanges, constants, degenerate cases, dependency cycles and statement–conclusion consistency.

Result accounting

Result	Disposition	Status
Proposition 1	Same-fibre witness proves necessity; identity map proves sufficiency.	PASS
Corollary 1	Rowwise inverse and zero-row boundary are exact.	PASS
FWL/design	Residualized equation, inverse bound and singular-direction claim are valid.	PASS
Joint ridge	Block elimination gives the exact Schur complement; positive penalty gives uniqueness.	PASS
Proposition 2	Telescoping and spectral bounds give the finite-horizon response-transfer inequality.	PASS

Proof-obligation closure

There are zero open FATAL or CRITICAL proof issues. All results state the finite-dimensional domains and the assumptions used at application. The proof contains no limits, asymptotic notation, expectations, integrals or probability-mode interchanges. The representation, design and propagation dependency branches are acyclic.

Counterexample pass

The red-team pass tested: factorization with unequal topologies; zero and nonzero exposure rows; collisions inside the diagonal class; individually informative but jointly collinear network lags; zero or collinear ridge designs; unstable companions; and bounded low-dimensional companions. No counterexample was found inside the declared hypotheses. Unstable companions are outside Proposition 2’s bounded-power assumption.

Scope boundary

The mathematical results do not establish CP rank recovery, CP-ALS convergence, estimator superiority, empirical bounded-power validity, bootstrap coverage or the validity of any archived RCEP/NYC number. R006e and R006f outcome stages remained closed and were neither inspected nor executed.

Audited source hashes

File	SHA-256 prefix
methods_theory.md	7679c3ccc2c8e8ba
supp_note1_notation.md	985c492e2b7f88da
supp_note3_propagation.md	e373666fe2a3b91b